Assignment 3

Problem 3.1

3×3 warehouse

Propositional symbols:

$D_{i,j}$: Damaged floor at i,j

$F_{i,j}$: Forklift at i,j

$P_{i,j}$: Package at i,j

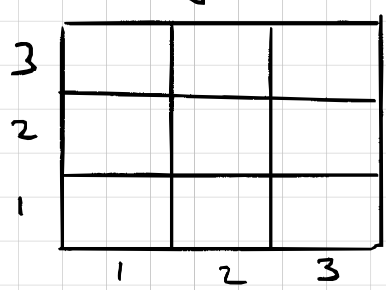
$C_{i,j}$: Creaking perceived at i,j

$R_{i,j}$: Rumbling perceived at i,j

$S_{i,j}$: Square (i,j) is safe

1) Encode Physics

- Creaking at a location iff damaged floor in an adjacent location.



$C_{1,1} \Leftrightarrow D_{1,2} \vee D_{21}$

$C_{1,2} \Leftrightarrow D_{1,3} \vee D_{2,2}$ because $S_{1,1}$

$C_{1,3} \Leftrightarrow D_{1,2} \vee D_{2,3}$

$C_{2,1} \Leftrightarrow D_{2,2} \vee D_{3,1}$ because $S_{1,1}$

$C_{2,2} \Leftrightarrow D_{1,2} \vee D_{2,3} \vee D_{2,1} \vee D_{3,2}$

$C_{2,3} \Leftrightarrow D_{2,2} \vee D_{1,3} \vee D_{3,3}$

$C_{3,1} \Leftrightarrow D_{2,1} \vee D_{3,3}$

$C_{3,2} \Leftrightarrow D_{3,3} \vee D_{2,2} \vee D_{3,1}$

$C_{3,3} \Leftrightarrow D_{2,3} \vee D_{3,2}$

- Rumbling at a location iff forklift in an adjacent square

Rewrite equations for damage floor because same conditions apply.

$R_{1,1} \Leftrightarrow F_{1,2} \vee F_{2,1}$
$R_{1,2} \Leftrightarrow F_{1,3} \vee F_{2,2}$ because $S_{1,1}$
$R_{1,3} \Leftrightarrow F_{1,2} \vee F_{2,3}$
$R_{2,1} \Leftrightarrow F_{2,2} \vee F_{3,1}$ because $S_{1,1}$
$R_{2,2} \Leftrightarrow F_{1,2} \vee F_{2,3} \vee F_{2,1} \vee F_{3,2}$
$R_{2,3} \Leftrightarrow F_{2,2} \vee F_{1,3} \vee F_{3,3}$
$R_{3,1} \Leftrightarrow F_{2,1} \vee F_{3,3}$
$R_{3,2} \Leftrightarrow F_{3,3} \vee F_{2,2} \vee F_{3,1}$
$R_{3,3} \Leftrightarrow F_{2,3} \vee F_{3,2}$

- A square is safe iff it has no damaged floor and no forklift.

$$S_{i,j} \Leftrightarrow \neg D_{i,j} \wedge \neg F_{i,j}$$
$$\Leftrightarrow \neg (D_{i,j} \vee F_{i,j})$$

2) Initial knowledge

- Starting square (1, 1) is safe.

$$S_{1,1} \Rightarrow \neg D_{1,1} \wedge \neg F_{1,1}$$
$$\Rightarrow \neg (D_{1,1} \vee F_{1,1})$$

- At least one square is damaged

$$D_{i,j} \vee \neg D_{i,j}, \alpha$$

- At least one square has a forklift

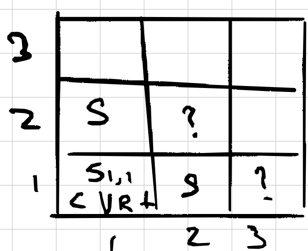
$$F_{i,j} \vee \neg F_{i,j}, \alpha$$

3) Scenario reasoning:

- Robot at (1, 1) and perceives no rumbling and no creaking.

$$C_{1,1} \vee R_{1,1} = \perp$$

- logical inference to determine which squares are provably safe.



If $C_{1,1} \vee R_{1,1} = \perp$

$$D_{1,2} \vee D_{2,1} \vee F_{1,2} \vee F_{2,1} = \perp$$

$$\neg (D_{1,2} \wedge D_{2,1} \wedge F_{1,2} \wedge F_{2,1})$$

$$S_{1,2} \Rightarrow \neg (D_{1,2} \wedge F_{1,2})$$
$$S_{2,1} \Rightarrow \neg (D_{2,1} \wedge F_{2,1})$$

4) Exploration: Robot moves to (2, 1) and perceives creaking but no rumbling.

New percepts $\begin{cases} C_{2,1} = T \\ R_{2,1} = \perp \end{cases}$

still unknown to be safe but no forklift $\leftarrow \begin{cases} F_{2,2} \vee F_{2,3} = \perp \\ D_{2,2} \vee D_{2,3} = T \end{cases}$

Known safe squares $\begin{cases} S_{1,1} \\ S_{2,1} \\ S_{1,2} \end{cases}$